BRAC University

Department of Electrical & Electronic Engineering

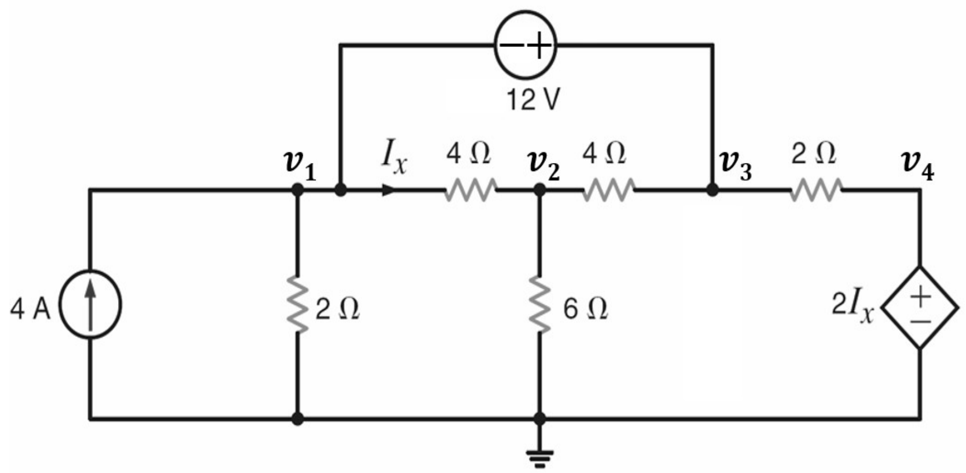
Assignment 1b, Spring 2025

EEE/ECE101: Electrical Circuits I

Total Marks: 120

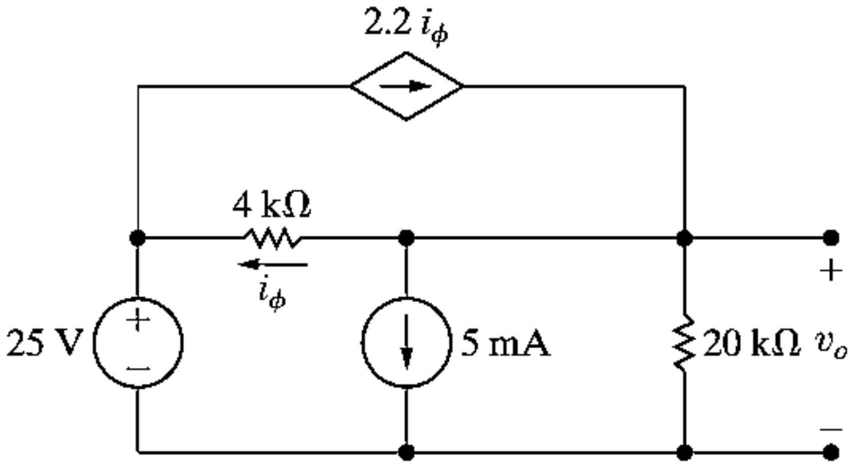
Q1. [10 marks]

Apply the **nodal analysis technique** to find the node voltages, v_1, v_2, v_3, and v_4 in the following circuit.

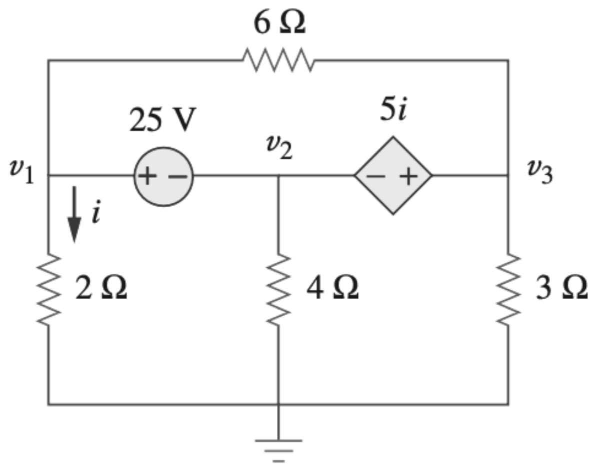


Q2. [10 marks]

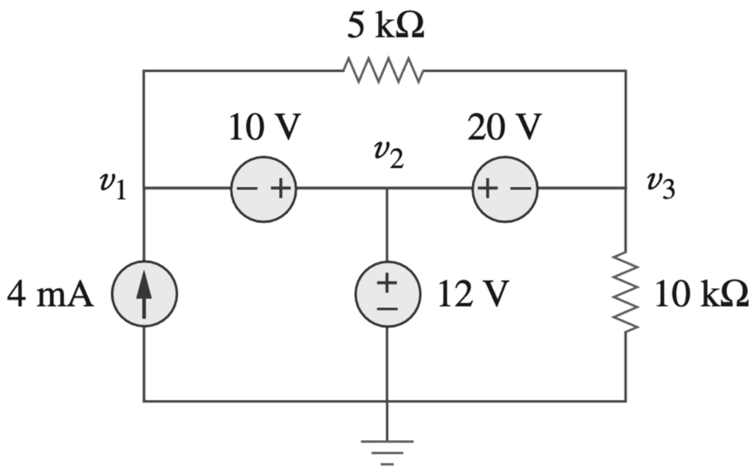
Apply the **nodal analysis technique** to find the voltage, v_0 in the following circuit.



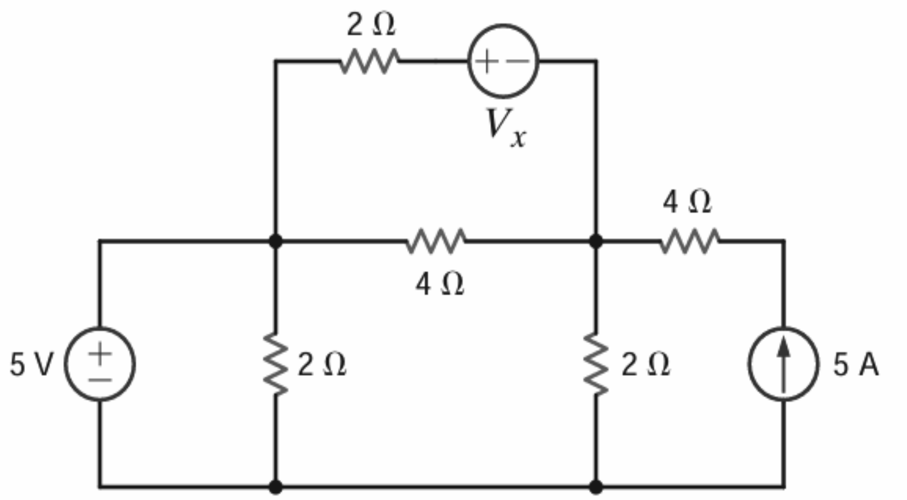
Q3. [10 marks]
Apply the **nodal analysis technique** to find the node voltages, v_1, v_2, and v_3 in the following circuit.



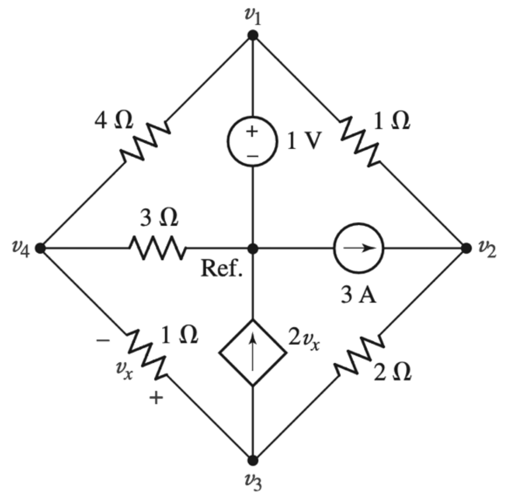
Q4. [10 marks]
Apply the **nodal analysis technique** to find the node voltages, v_1, v_2, and v_3 in the following circuit.



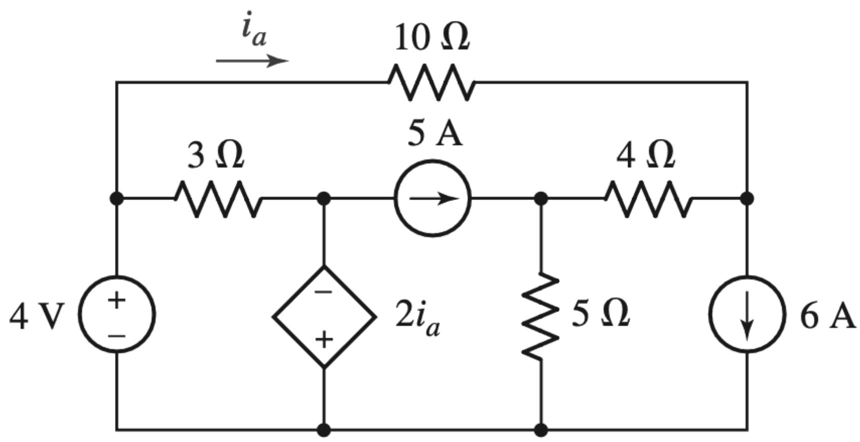
Q5. [10 marks]
Using **nodal analysis**, find the value of V_x, such that the 5-A current source supplies 50 W.



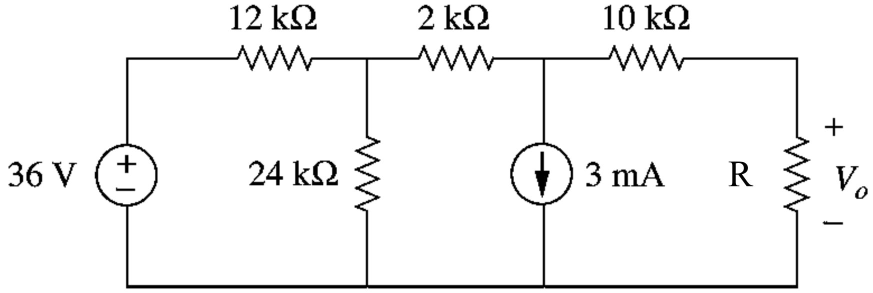
Q6. [10 marks]
Determine all four nodal voltages for the following circuit.



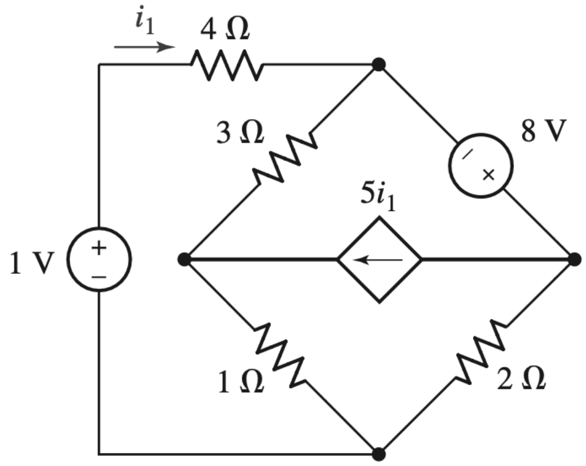
Q7. [10 marks]
Using **mesh analysis**, determine the power absorbed by the 10-ohm resistor.



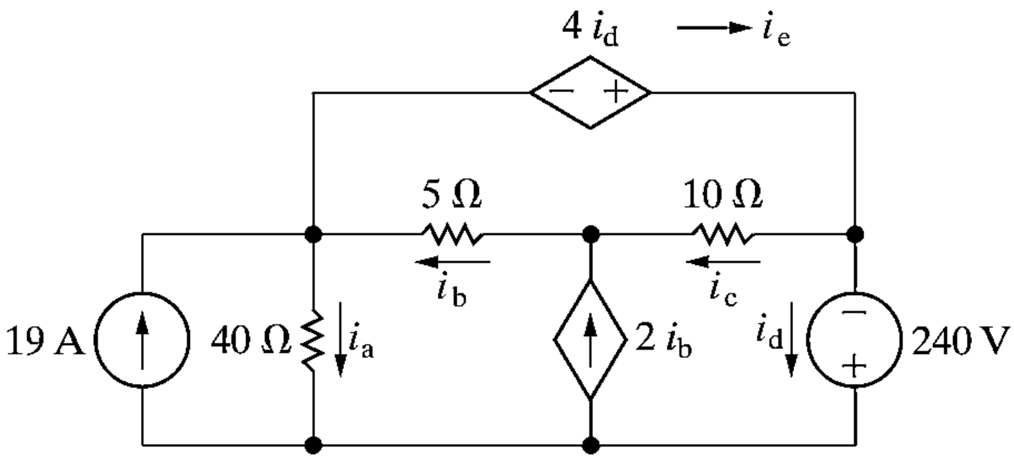
Q8. [10 marks]
Using **mesh analysis**, find the value of the resistance, R such that the voltage, $V_0 = -1\ V$, for the following circuit.



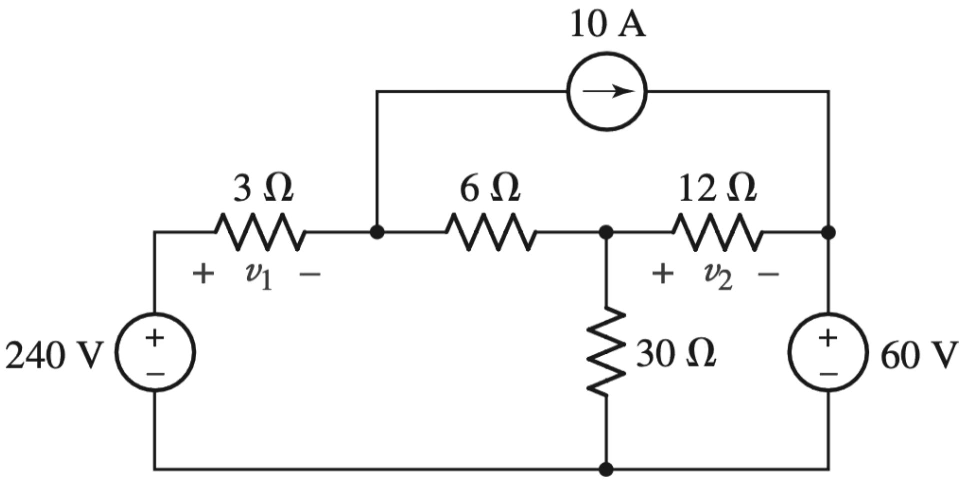
Q9. [10 marks]
Using **mesh analysis**, determine the power supplied by the 1V source.



Q10. [10 marks]
Find the branch currents, i_a, i_b, i_c, i_d, i_e for the following circuit using the **mesh analysis technique.**

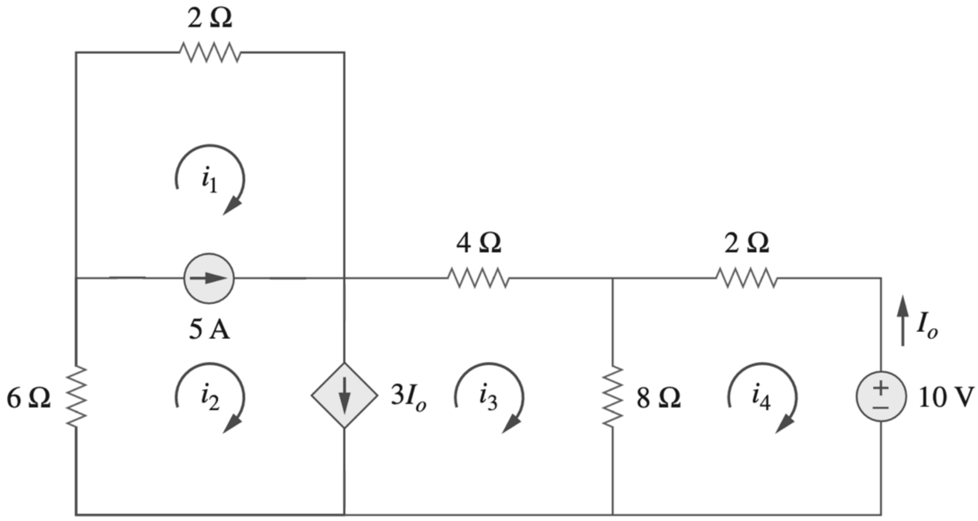


Q11. [10 marks]
Using **mesh analysis**, determine the voltages v_1 and v_2.



Q12. [10 marks]

For the following circuit, determine i_1, i_2, i_3, and i_4 using mesh analysis.



EEE 101

Assignment 1b

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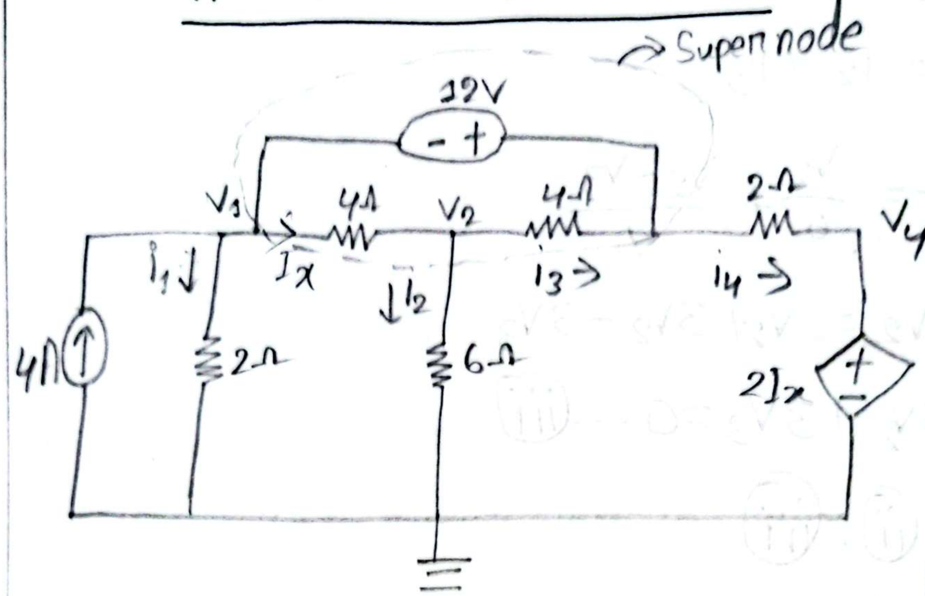
Ans to the Q. No-1



$i_1 = \frac{V_1}{2}$

$i_2 = \frac{V_2}{6}$

$i_3 = \frac{V_2 - V_3}{4}$

$i_4 = \frac{V_3 - V_4}{2}$

$I_x = \frac{V_1 - V_2}{4}$

$\therefore V_4 = 2I_x$

$\Rightarrow V_4 = \frac{V_1 - V_2}{2}$

Applying KCL at Supernode,

$4 + i_3 = i_1 + i_2 + i_4$

$\Rightarrow 4 + \frac{V_2 - V_3}{4} = \frac{V_1}{2} + \frac{V_2}{6} + \frac{V_3 - V_4}{2}$

$\Rightarrow 3V_1 - 2V_2 + 3V_3 - 2V_4 = 16$

$\Rightarrow 3V_1 - 2V_2 + 3V_3 - 2 \times \left(\frac{V_1 - V_2}{2}\right) = 16$

$\therefore 2V_1 - V_2 + 3V_3 = 16 \quad \ldots \text{(i)}$

Applying KVL at Supernode,

$-V_1 + V_3 - 12 = 0$

$\therefore V_3 - V_1 = 12 \quad \ldots \text{(ii)}$

Applying KCL at node V_2,

$$I_x = i_2 + i_3$$

$$\Rightarrow \frac{V_1 - V_2}{4} = \frac{V_2}{6} + \frac{V_2 - V_3}{4}$$

$$\Rightarrow 3V_1 - 3V_2 = 2V_2 + 3V_2 - 3V_3$$

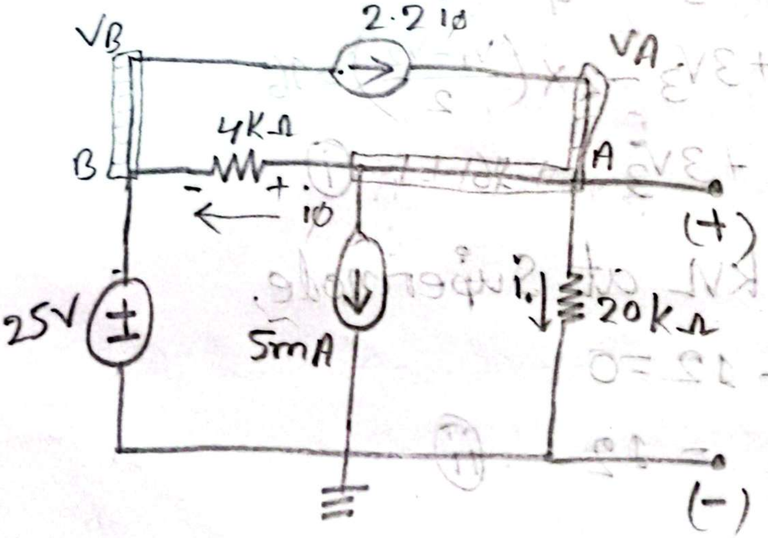
$$\Rightarrow 3V_1 - 8V_2 + 3V_3 = 0 \quad \cdots \text{(iii)}$$

from (i), (ii), (iii)

$V_1 = -3.65V$, $V_2 = 1.76\ V$, $V_3 = 8.35V$;

$$V_4 = \frac{V_1 - V_2}{2} = \frac{-3.65 - 1.76}{2} = -2.71\ V$$ Ans

Answer to the Q. No-2



$$i_1 = \frac{V_o}{20}$$

$$i_o = \frac{V_o - V_B}{4} = \frac{V_o - 25}{4}$$

$V_B = 25V$

Applying KCL at node A ⇒

$i_o + i_\phi + 5 = 2.2 i_\phi$

$\Rightarrow \frac{V_o}{20} + \frac{V_o - 25}{4} + 5 = 2.2 \times \frac{V_o - 25}{4}$

$\Rightarrow \frac{V_o + 5V_o - 125 + 100}{20} = \frac{2.2V_o - 55}{4}$

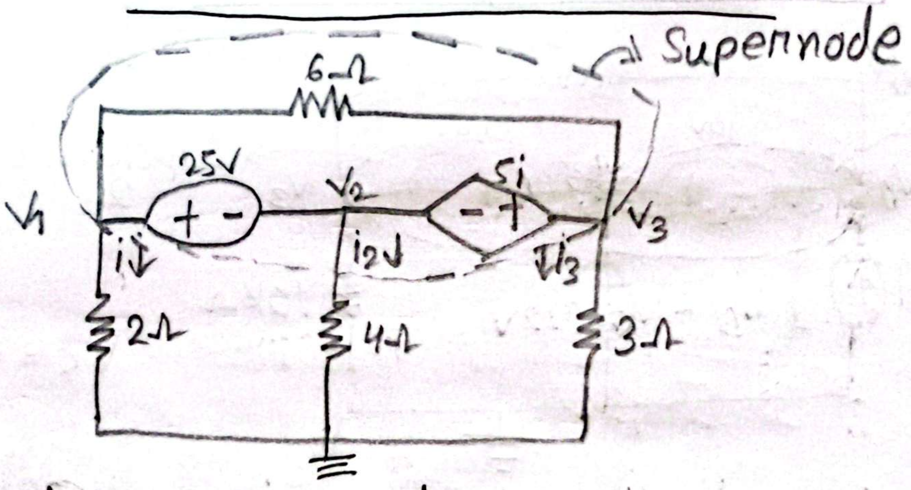
$\Rightarrow 24V_o - 100 = 44V_o - 1100$

$\Rightarrow 44V_o - 24V_o = 1100 - 100$

$\Rightarrow 20V_o = 1000$

$V_o = 50V$ Ans.

Answer to the Q. NO-3



Applying KCL at Supernode,

$i + i_2 + i_3 = 0$

$\Rightarrow \frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0$

$\therefore 6V_1 + 3V_2 + 4V_3 = 0$ ...... (i) again, $V_1 - V_2 = 25$ ... (ii)

again, $V_2 - V_3 = -5i$

$\Rightarrow V_2 - V_3 = -\frac{5V_1}{2}$

$\Rightarrow 2V_2 - 2V_3 + 5V_1 = 0$ ... (iii)

from (i), (ii), (iii) $\Rightarrow$

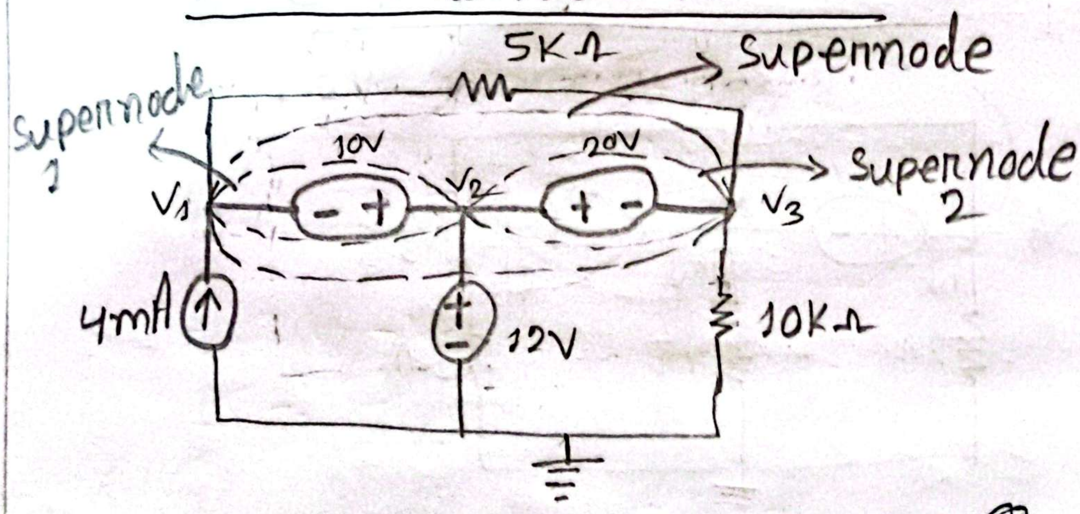
$V_1 = 7.61V$

$V_2 = -17.39V$

$V_3 = 1.63V$

Ans

Answer to the Q. NO-4



From, Supernode-1, $V_2 - V_1 = 10$ ... (i)

From Supernode-2, $V_3 - V_2 = 20$ ... (ii)

again,

$V_2 - 0 = 12$

$V_2 = 12V$

from (i) ⇒ $12 - V_1 = 10$

$\therefore V_1 = 2V$

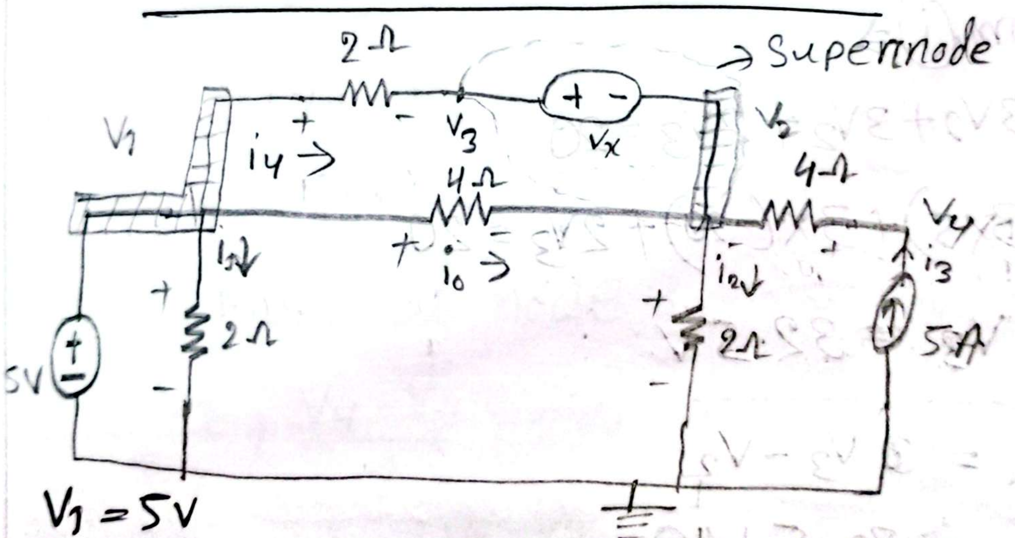
from (ii) ⇒

$12 - V_3 = 20$

$\therefore V_3 = -8V$

$$\therefore \left.\begin{aligned} V_1 &= 2V \\ V_2 &= 12V \\ V_3 &= -8V \end{aligned}\right\} \text{Ans}$$

Answer to the Q. NO-5



$V_1 = 5V$

Applying KCL at supernode (V_2, V_x, V_3)

$$i_3 + i_0 = i_2 + i_4$$

$$\Rightarrow 5 + \frac{V_1 - V_2}{4} = \frac{V_2}{2} + \frac{V_3 - V_1}{2}$$

$$\Rightarrow (20 + V_1 - V_2)2 = (V_2 + V_3 - V_1) \times 4$$

$\Rightarrow (20+V_1-V_2) = (V_2+V_3-V_1) \times 2$

$\Rightarrow 20+V_1-V_2 = 2V_2+2V_3-2V_1$

$\Rightarrow -3V_1+3V_2+2V_3 = 20 \quad \ldots (i)$

now,

$P_4 = V_4 I$

$\Rightarrow V_4 = \frac{50}{5} = 10V.$

$\frac{V_4-V_2}{4} = 5$

$\Rightarrow 10-V_2 = 20$

$\Rightarrow V_2 = -10V$

again,

$V_x = V_3 - V_2$

from (i) $\Rightarrow$

$-3V_1+3V_2+2V_3 = 20$

$\Rightarrow (-3\times5)+3\times(-10)+2V_3 = 20$

$\Rightarrow V_3 = 32.5V$

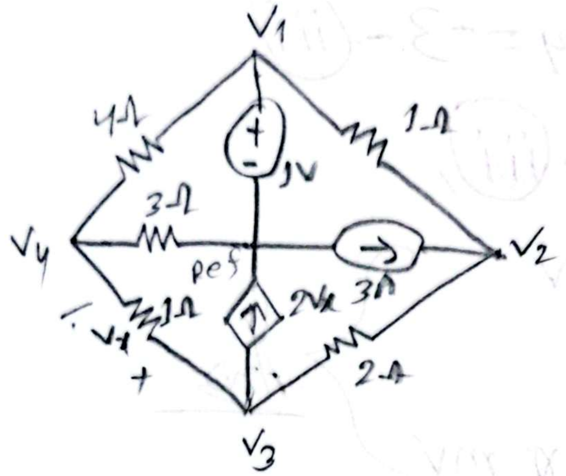
$\therefore V_x = \cancel{3}\, V_3 - V_2$

$= 32.5+10$

$= 42.5V$

Ans

Answer to the Q. No-6



Applying KCL at node-2,

$$\frac{V_2-V_1}{1}+\frac{V_2-V_3}{2}=3$$

$$\Rightarrow 2V_2-2+V_2-V_3=6$$

$$\Rightarrow 3V_2-V_3=8 \quad \cdots (i)$$

Applying KCL at node-3,

$$\frac{V_2-V_3}{2}+\frac{V_4-V_3}{1}=2V_x$$

$$\Rightarrow 2(V_3-V_4)=\frac{V_2-V_3}{2}+V_4-V_3$$

$$\rightarrow V_2-76V_3+6V_4=0 \quad \cdots (ii)$$

Applying KCL at node-4,

$$\frac{V_4-V_1}{4}+\frac{V_4}{3}+\frac{V_4-V_3}{1}=0$$

$\Rightarrow 3V_4 - 3V_3 + 4V_4 + 12V_4 - 12V_3 = 0$

$\Rightarrow 12V_3 - 19V_4 = -3$ --- (iii)

from, (i), (ii), (iii),

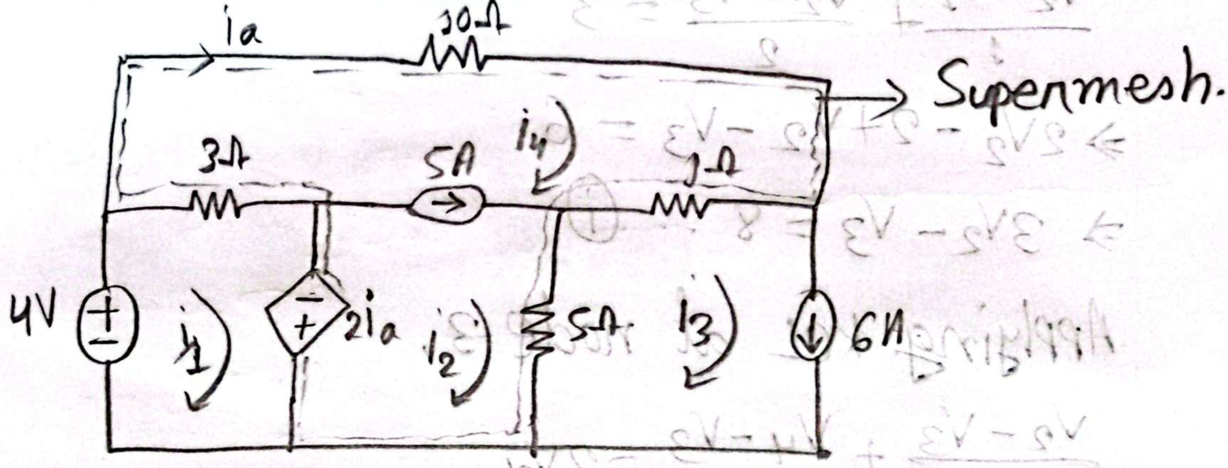
$V_2 = 3.085V$

$V_3 = 2.256\ V$

$V_4 = 951.219\cancel{8}\ mV$

Ans

Answer to the Q. NO-7



Applying ~~KCL~~ KVL on ~~meshes~~ Supermesh,

~~10i4~~

$10i_4 + 4(i_4 - i_3) + 5(i_2 - i_3) + 2i_a + 3(i_4 - i_1) = 0$

$\Rightarrow 10i_4 + 4i_4 - 4i_3 + 5i_2 - 5i_3 + 2i_4 + 3i_4 - 3i_1 = 0$ $[i_4 = i_a]$

$\therefore -3i_1 + 5i_2 + 19i_4 = 54$ --- (i)

Applying KCL on Supermesh,

$$i_2 - i_4 = 5 \ldots \text{(ii)}$$

Applying KVL on mesh-1,

$$3(i_1 - i_4) - 2i_a - 4 = 0$$

$$\Rightarrow 3i_1 - 3i_4 - 2i_4 - 4 = 0 \quad [i_a = i_4]$$

$$\therefore 3i_1 - 5i_4 = 4 \ldots \text{(iii)}$$

Applying KVL on mesh-4,

$$i_4 = -5A.$$

from, (i), (ii), (iii) $\Rightarrow$

$$i_1 = 4.23A$$

$$i_2 = 6.74A$$

$$i_4 = 1.74A.$$

$$\therefore i_a = i_4 = 1.74A.$$

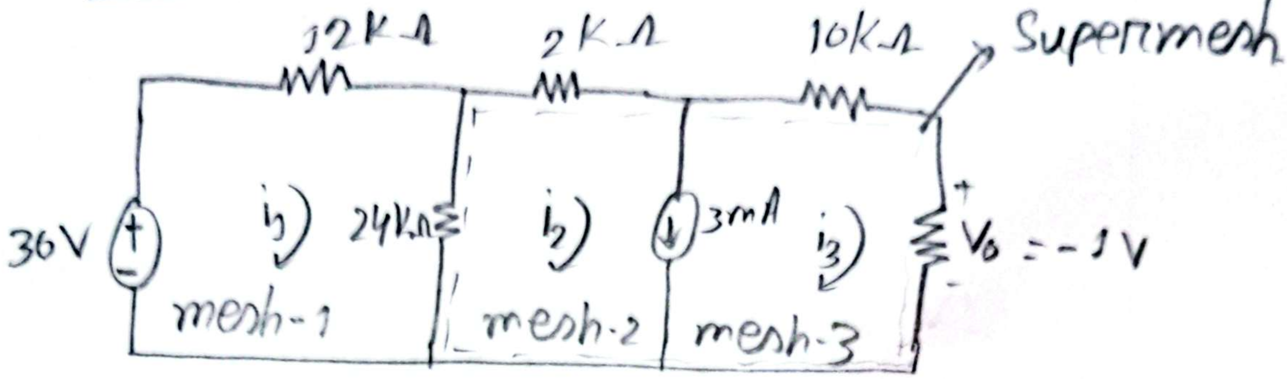
$\therefore$ Power absorbed by 10Ω resistor, $P = i_a^2 \times R$

$$= (1.74)^2 \times 10$$

$$= 30.1662\,W$$

Ans

Answer to the Q. NO-8



Applying KVL on Supermesh,

$$24(i_2 - i_1) + 2i_2 + 10i_3 - 1 = 0$$

$$\Rightarrow -24i_1 + 26i_2 + 10i_3 = 1 \quad \cdots \text{(i)}$$

Applying KCL on Supermesh,

$$i_2 - i_3 = 3 \quad \cdots \text{(ii)}$$

Applying KVL on mesh-1,

$$\cancel{10i}\ 12i_1 + 24(i_1 - i_2) - 36 = 0$$

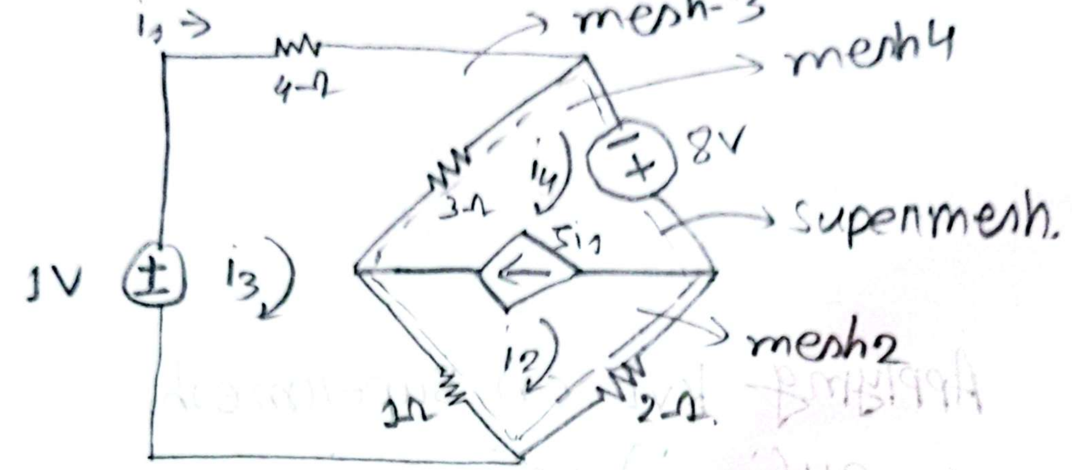
$$\Rightarrow 36i_1 - 24i_2 = 36 \quad \cdots \text{(iii)}$$

from, (i), (ii), (iii) →

$$i_1 = 2.83A, \quad i_2 = 2.75A, \quad i_3 = -0.25A.$$

$$R = \frac{V_0}{i_3} = \frac{-1}{i_3} = \frac{-1}{-0.25} = 4K\Omega \quad \text{Ans}$$

Answer to the Q. NO-9



Applying KVL on supermesh,

$3(i_4 - i_3) - 8 + 2i_2 + 1(i_2 - i_3) = 0$

$\Rightarrow 3i_4 + 3i_2 - 4i_3 = 8 \quad \ldots (i)$

Applying KCL on supermesh,

$i_4 - i_2 = 5i_1$

$\Rightarrow i_4 - i_2 = 5i_3 \quad [i_1 = i_3]$

$\therefore i_4 - i_2 - 5i_3 = 0 \quad \ldots (ii)$

Applying KVL on mesh3,

$-1 + 4i_3 + 3(i_3 - i_4) + 1(i_3 - i_2) = 0$

$\Rightarrow -3i_4 - i_2 + 8i_3 = 1 \quad \ldots (iii)$

from, (i), (ii), (iii)

$i_2 = -33.5A$

$i_3 = 19A$

$i_4 = 61.5A$

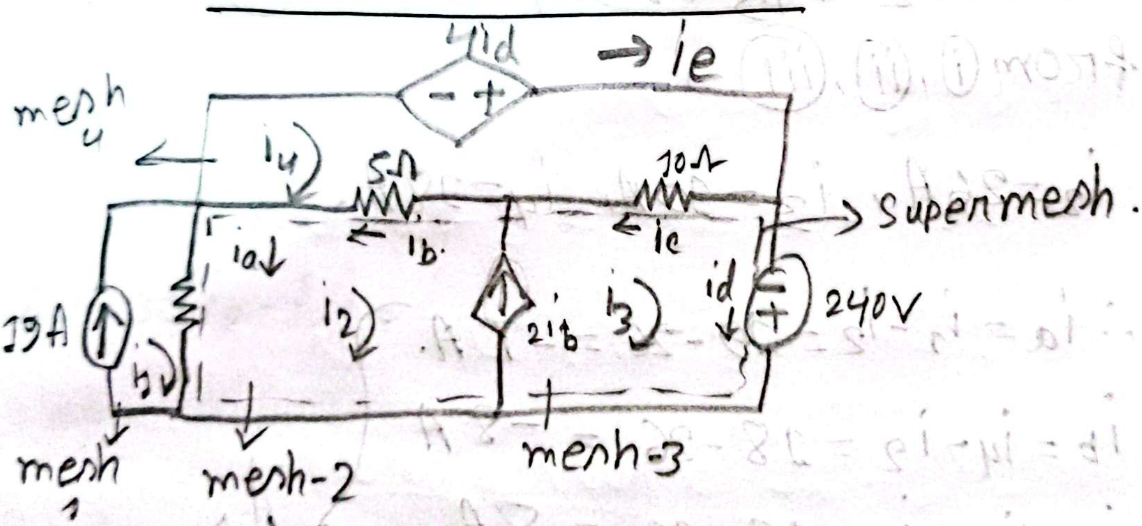
The power supplied by the 1V source is

$P = Vi_3$

$= 1 \times 19$

$= 19\ W$ Ans

Ans to the Q. No - 10



Applying KVL on supermesh,

$40(i_2 - i_1) + 5(i_2 - i_4) + 10(i_3 - i_4) - 240 = 0$

$\Rightarrow -40i_1 + 45i_2 + 10i_3 - 15i_4 = 240$

$\Rightarrow 45i_2 + 10i_3 - 15i_4 = 1000$ .. (i) $[i_1 = 19A]$

Applying KCL on Supermesh.

$i_3 - i_2 = 2i_b$

$\Rightarrow i_3 - i_2 = 2(i_4 - i_2)$ $[i_b = i_4 - i_2]$

$\Rightarrow -i_2 - i_3 + 2i_4 = 0 \cdots$ (ii)

Applying KVL on mesh-4.

$-4i_d + 10(i_4 - i_3) + 5(i_4 - i_2) = 0$

$\Rightarrow -4i_3 + 10i_4 - 10i_3 + 5i_4 - 5i_2 = 0,$ $[i_d = i_3]$

$\Rightarrow \cancel{8} - 5i_2 - 14i_3 + 15i_4 = 0 \cdots$ (iii)

from (i), (ii), (iii) $\Rightarrow$

$i_2 = 26A,\ i_3 = 10A,\ i_4 = 18A.$

$\therefore i_a = i_1 - i_2 = 19 - 26 = -7\ A.$

$i_b = i_4 - i_2 = 18 - 26 = -8\ A$

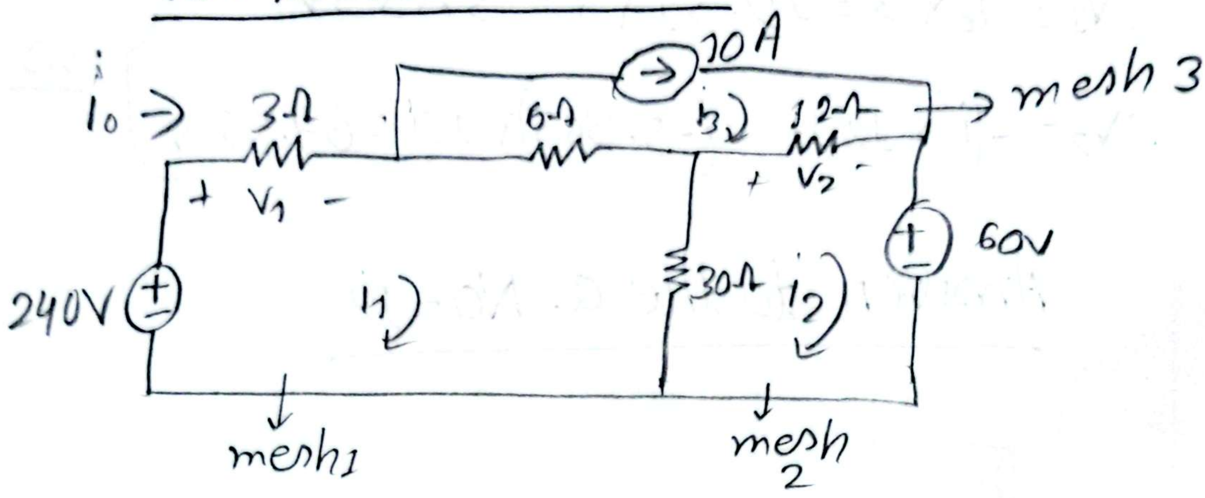
$i_c = i_4 - i_3 = 18 - 10 = 8A$

$i_d = i_3 = 10A$

$i_e = i_4 = 18A$

Ans

Ans to the Q. NO-11



Applying KVL on mesh 1,

$-240+3i_1+6(i_1-i_3)+30(i_1-i_2)=0$

$\Rightarrow -240+3i_1+6i_1-(6\times10)+30i_1-30i_2=0$

$\Rightarrow 39i_1-30i_2=300$ ...... (i)

Applying KVL on mesh-2

$60+30(i_2-i_1)+12(i_2-i_3)=0$

$\Rightarrow 60+30i_2-30i_1+12i_2-12\times10=0$

$\Rightarrow 42i_2-30i_1=60$ ..... (ii)

from,

$i_1=19.591\ A,\ i_2=15.365\ A.$

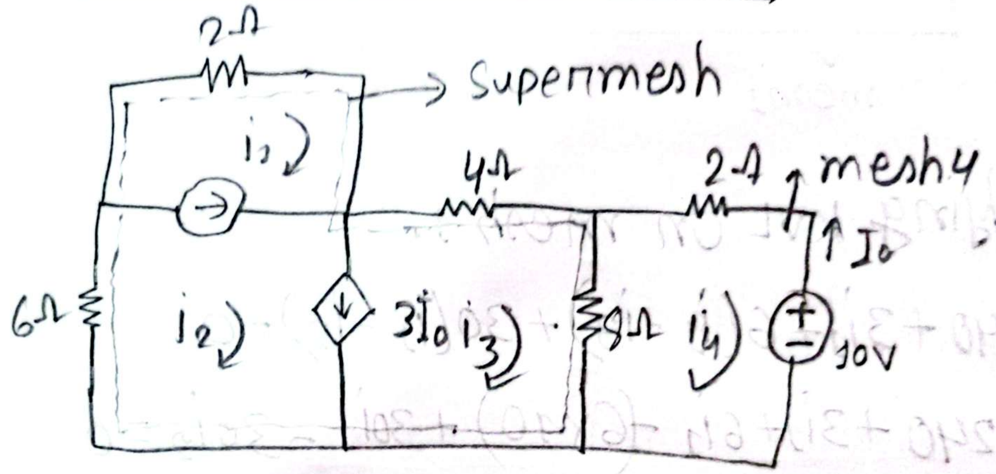
$i_0=i_1=19.59A,\ i'=i_3-i_2=(10-15.365)A$

$=-5.365\ A$

$\therefore V_1 = I_0 \times 3 = 19.51 \times 3 = 58.53V$

$\therefore V_2 = -(i' \times 12) = -(-5.365) \times 12 = 64.38\ V$ Ans

Answer to the Q. No-12



Applying KVL around Supermesh,

$6i_2 + 2i_1 + 4i_3 + 8(i_3 - i_4) = 0$

$\Rightarrow i_1 + 3i_2 + 6i_3 - 4i_4 = 0$ --- (i)

Applying KVL around mesh-4,

$8(i_4 - i_3) + 2i_4 + 10 = 0$

$\Rightarrow 5i_4 - 4i_3 = -5$ ---- (ii)

Applying KCL on Supermesh,

$$i_2 - i_3 = 3I_0$$

$$\Rightarrow i_2 - i_3 = -3i_4 \quad [I_0 = -i_4]$$

$$\Rightarrow i_2 - i_3 + 3i_4 = 0 \quad \cdots \text{(iii)}$$

Applying KCL,

$$i_2 - i_1 = 5 \quad \cdots \text{(iv)}$$

from,

(i), (ii), (iii), (iv)

$i_1 = -7.5A$, $i_2 = -2.5A$, $i_3 = 3.93A$, $i_4 = 2.143A$

Ans